Emergent Partitioning in Continuous Predictive Systems:

A Variance- and Cost-Constrained Alternative to Explicit Gating

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Author Statement

This work represents a structured synthesis of observable dynamics under constraint-based reasoning. It is presented as a falsifiable model intended for independent validation, simulation, and empirical testing. No claim of final authority is made. The framework remains open to correction, refinement, and extension under lawful observation.

Abstract

Partitioning in predictive systems is typically implemented via explicit gating mechanisms (e.g., Mixture-of-Experts). This work demonstrates that partitioning can instead emerge from continuous, variance- and cost-constrained error minimization within a shared state-space.

Competing priors remain jointly active and are continuously weighted. Under sustained structured error, the system exhibits clustering, precision-weight collapse, and context-dependent separation, producing soft, reversible partitions without predefined boundaries.

The framework predicts smooth weight transitions, history-dependent clustering, and gradual partition formation, in contrast to the threshold discontinuities of discrete gating.

A minimal empirical protocol using multi-cue perceptual decision tasks is defined, with clear observables and falsifiability criteria.

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1. Introduction

Predictive systems must resolve competition among candidate priors. Standard approaches use explicit gating to enforce specialization via discrete selection.

These approaches impose:

- fixed partition boundaries,
- threshold-based switching,

- externally defined structure.

We instead test:

Partitioning emerges from constrained error minimization, rather than being imposed.

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2. Formal Framework

2.1 Continuous Arbitration

$$\hat{y} = \sum_i w_i P_i(x), \quad \sum_i w_i = 1$$

$$\mathcal{L} = \mathbb{E}[(y - \hat{y})^2] + \lambda C(w)$$

2.2 Governing Constraints

1. Error minimization
2. Variance sensitivity
3. Cost constraint

No gating or expert assignment is assumed.

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3. Emergent Partitioning Dynamics

Partitioning arises from:

- **Variance clustering** → coherent error regions
- **Precision collapse** → local specialization
- **Decoherence** → separation across contexts
- **Temporal persistence** → stabilization filter
- **Cost constraint** → minimal sufficient structure

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4. Continuous Arbitration

- Outputs remain **weighted composites**
- No discrete switching
- Poor priors undergo **gradual, proportional demotion**

5. Comparison to MoE

Property	Continuous	MoE
Selection	Continuous	Discrete
Boundaries	Emergent	Fixed
Transitions	Smooth	Threshold
Adaptation	History-based	Reset-like
Structure	Learned	Imposed

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6. Empirical Protocol (v0.1)

Task

Multi-cue perceptual decision with:

- overlapping cues
- reliability control
- conflict
- context shifts

Conditions

- Baseline
- Conflict
- Reliability shift
- Context shift
- High variance

Observables

- Accuracy
- Reaction time
- Adaptation curves
- Weight inference
- Variance structure

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7. Falsifiable Predictions

Continuous model:

- smooth weight shifts
- gradual clustering
- reversible partitions

MoE:

- threshold switching
- abrupt transitions

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8. Separation Criteria

- Weight continuity
- Transition signatures
- Mixed-cue behavior
- Adaptation curves
- Variance structure
- RT / cost

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9. Early Partition Formation

Under high variance:

- initial weighting remains diffuse
- transient clusters form
- only persistent clusters stabilize
- early partitions remain reversible

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10. Discussion

Partitioning, specialization, and arbitration are unified as consequences of:

error minimization under variance and cost constraints

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11. Conclusion

Partitioning is not required, it is **discovered**.

Continuous systems naturally produce:

- adaptive specialization
- soft partitioning
- stable arbitration

without explicit gating.

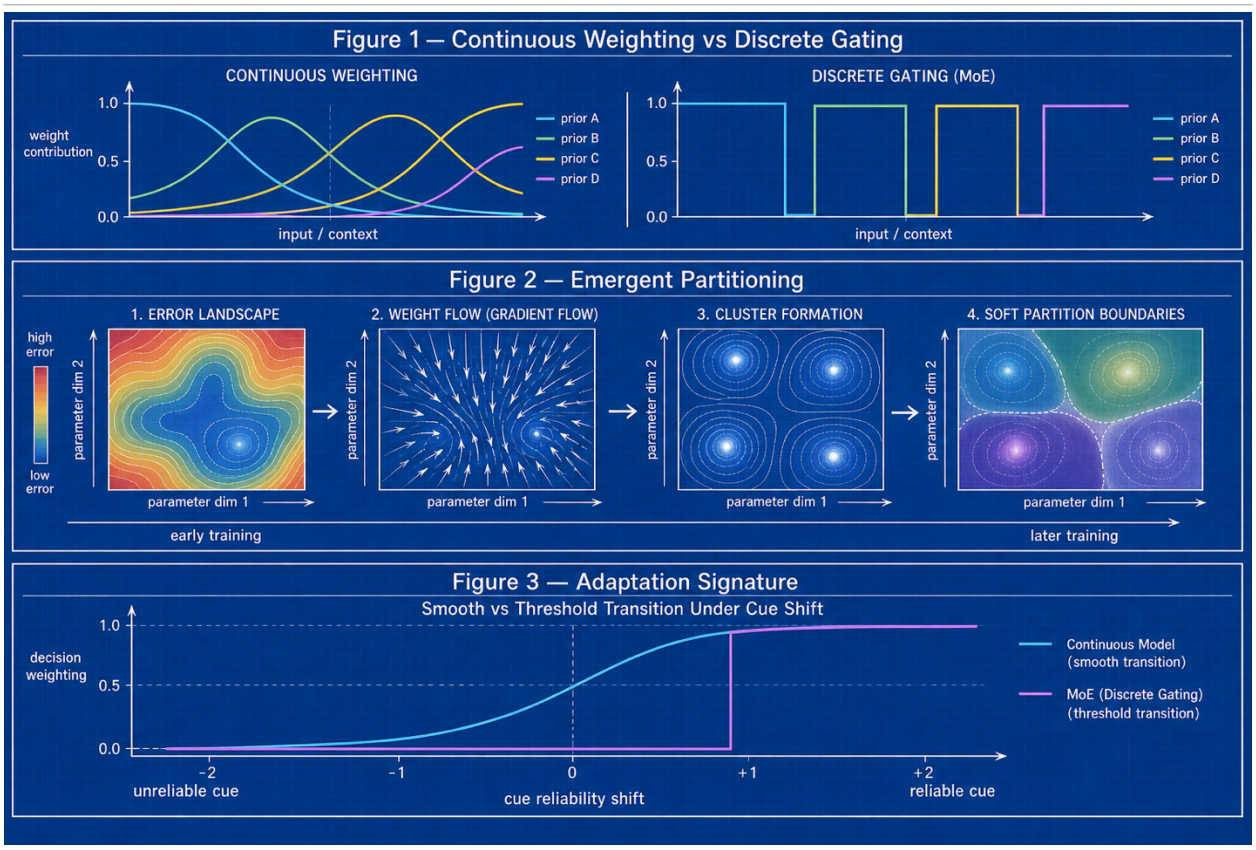

Figure 1 — Continuous Weighting vs Discrete Gating

Smooth overlapping weight curves vs step-function assignment.

Figure 2 — Emergent Partitioning

Error landscape → gradient flow → clustering → soft partitions.

Figure 3 — Adaptation Signature

Smooth vs threshold transition under cue reliability shift.

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